MAR BASELIOS INSTITUTE OF TECHNOLOGY AND SCIENCE

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DEPARTMENT
OF
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Smart College Id Based Attendance System

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CONTENTS

- Introduction
- Objectives
- Timeline & Milestones
- Implementation
- Demonstration
- Problem-Solving
- Innovation
- Creativity
- Quality
- Contributions
- Conclusion
- Future Scope

INTRODUCTION

Overview of the Problem

- Traditional attendance systems use manual roll calls or paper registers.
- Time-consuming and prone to human errors.
- Risk of proxy attendance by students.
- Difficult to track students' entry and exit times accurately.
- Requires considerable effort to maintain attendance records.
- Need for an automated, reliable, and efficient attendance management system.

Why This Project?

- To simplify and automate the attendance process using RFID technology.
- To reduce manual effort and save time.
- To identify students using RFID cards.
- To display student details and attendance status on an LCD screen.
- To provide instant feedback using an LED and buzzer.
- To record students’ entry and exit times.
- To improve attendance accuracy and efficiency.

Project Objectives

- To develop an automated attendance system using Arduino Uno and RFID technology.
- To identify students by matching scanned RFID card UIDs with registered details.
- To indicate student entry and exit using LEDs and a buzzer.
- To display student names and attendance status on an LCD.
- To obtain and display the current date and time using a DS3231 RTC module.
- To reject unregistered cards and provide a simple, low-cost attendance solution.

Timeline & Milestones

Week 1 – Planning & Component Selection

- Identified attendance system problems and defined objectives.
- Selected Arduino UNO, RFID reader, LCD, cards, buzzer, and LEDs.

Week 2 – Circuit Assembly

- Connected hardware components.
- Completed the circuit setup.

Week 3 – Programming & Integration

- Developed Arduino code for RFID scanning and attendance tracking.
- Integrated LCD, buzzer, LEDs, and RTC module.

Week 4 – Testing & Final Implementation

- Tested RFID detection, entry/exit tracking, and hardware functionality.
- Fixed errors and completed project documentation.

Implementation

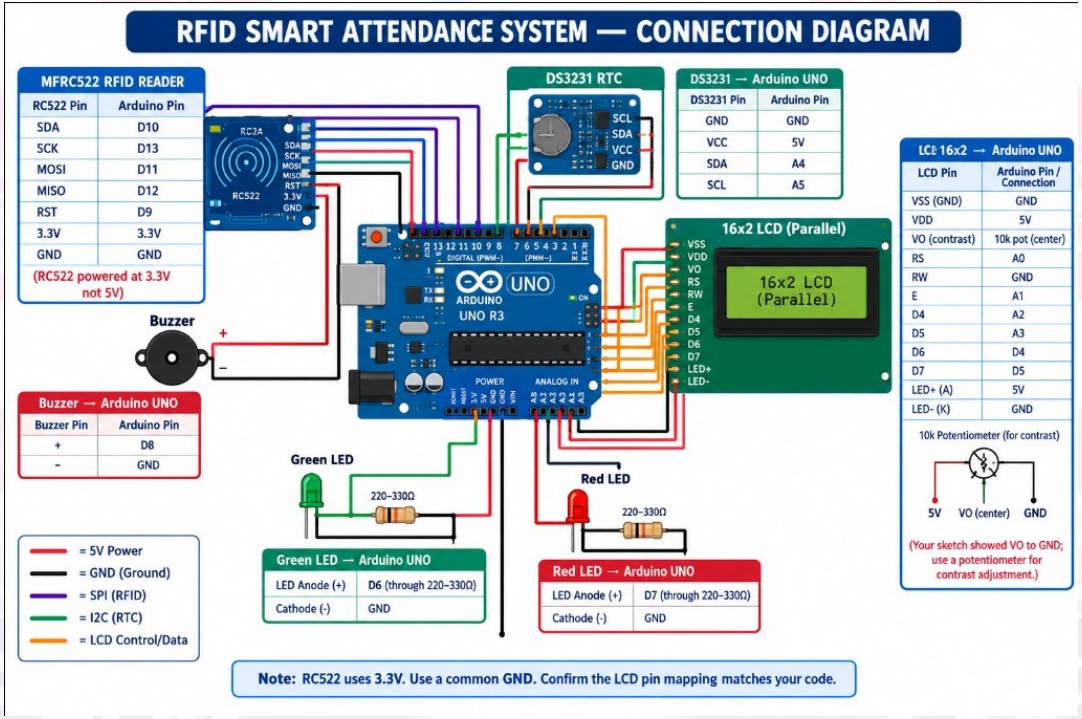
Program Description

- To develop the program using Arduino IDE and Arduino UNO.
- To read RFID card UID using the RC522 reader.
- To identify students by comparing scanned UIDs with stored IDs.
- To display student names and attendance status on the LCD.
- To provide scan confirmation using an LED and buzzer.
- To track student entry and exit through repeated scans.
- To automate attendance recording and reduce manual effort.

Steps Followed

Step 1: Identified problems in manual attendance systems.
Step 2: Selected Arduino UNO, RFID RC522, cards, LCD, buzzer, LEDs, and RTC.
Step 3: Connected all components directly to the Arduino UNO using appropriate pins.
Step 4: Developed code for RFID scanning and student identification.
Step 5: Displayed student details and attendance status on the LCD.
Step 6: Implemented LED and buzzer feedback.
Step 7: Tested card recognition and attendance tracking.
Step 8: Integrated components and verified system operation.

Demonstration



Problem-Solving

Challenges Faced

- **Arduino Upload Errors**: Faced communication errors while uploading code to Arduino UNO.
- **LCD Display Issues**: Initially, the LCD displayed a blank or blue screen.
- **RFID Recognition**: Ensuring correct card UID detection and student identification.
- **Hardware Integration**: Connecting and configuring all components correctly.
- **Program Debugging**: Fixing coding errors and ensuring accurate attendance and entry/exit tracking.

Solutions Implemented

- **Upload Troubleshooting**: Checked USB cable connections, COM port, and Arduino board settings.
- **LCD Configuration**: Verified LCD wiring, contrast settings, pin connections, and library compatibility.
- **UID Verification**: Used the Arduino Serial Monitor to read RFID UIDs and update the program.
- **Circuit Verification**: Checked component connections, power supply, and pin assignments.
- **Code Optimization**: Fixed coding errors and improved student identification, attendance tracking, and buzzer/LED feedback.

Innovation

Unique Features

- **RFID-Based Identification**: Automatically identifies students using unique RFID card IDs.
- **Automated Attendance Tracking**: Reduces manual effort and saves time during attendance marking.
- **Entry and Exit Monitoring**: Tracks whether a student is entering or leaving based on repeated card scans.
- **Real-Time LCD Display**: Displays the student's name and current attendance status instantly.
- **Audio and Visual Alerts**: Uses a buzzer and LED indicators to confirm card scans.
- **Expandable Design**: Can be extended to support more students and additional attendance management features.

Novel Ideas

Novel Ideas

Smart Entry–Exit Detection: Uses repeated RFID scans to distinguish between student entry and exit.

Real-Time Feedback System: Combines LCD messages, sound alerts, and LED signals for clear user interaction.

Low-Cost Automation: Uses affordable, widely available components to create an accessible attendance solution.

Future RTC Integration: Can record attendance timestamps to maintain time-based entry and exit records.

Future Data Storage: Can be enhanced with an SD card, database, or cloud platform to store and analyze attendance records.

Creativity

Approach

- **Input:** To scan RFID cards using the RC522 reader.
- **Processing**: To read the card UID and match it with stored student records using Arduino UNO.
- **Attendance Management**: To update student status as present or exiting based on previous scans.
- **Output**: To display student details on the LCD and provide feedback through the buzzer and LED.
- **Testing**: To verify RFID recognition, attendance tracking, and proper functioning of all components.

Optimization Techniques

- **Efficient UID Matching**: Uses predefined RFID card IDs for quick student identification.
- **Simplified Program Logic**: Uses conditional statements to manage entry and exit status.
- **Reduced Processing**: Avoids unnecessary calculations during card scanning.
- **Clear User Feedback**: Uses LCD messages, buzzer sounds, and LED indicators.
- **Modular Code Structure**: Separates functions for valid scans, invalid cards, and LED blinking.
- **Cost & Power Efficiency**: Uses a low-cost Arduino-based design with minimal additional hardware.

Quality

Accuracy

- Identifies students by matching scanned RFID UIDs with predefined student IDs.
- Displays the correct student name for recognized cards.
- Prevents incorrect identification by checking for invalid or unregistered cards.
- Updates entry/exit status based on the student's previous scan.

Efficiency

- Reduces time and effort required for manual attendance.
- Uses simple conditional logic to process card scans quickly.
- Employs low-cost hardware with minimal additional components.

Reliability

- Provides immediate feedback through the LCD, buzzer, and LEDs.
- Uses modular code to simplify debugging and maintenance.
- Ensures consistent operation through hardware testing and program debugging.

Contributions

- **Antony Augustine -** System Integration and Debugging : Responsible for integrating hardware and software, troubleshooting technical issues, and ensuring proper communication between components.
- **Dhananjai D** - Hardware Design and Circuit Assembly : Responsible for connecting the Arduino UNO, RFID reader, LCD display, buzzer, LEDs, and other hardware components.
- **Diya Dany** - Software Development and Programming : Responsible for writing the Arduino code for RFID card identification, student recognition, and attendance tracking.
- **Eldhose Joseph** – Hardware,Testing, Documentation and Presentation : Responsible for testing system accuracy and reliability, documenting project results, and preparing the final project report and presentation.

Conclusion

Key Learnings

- Gained practical experience in Arduino UNO programming and hardware interfacing.
- Learned how RFID technology enables contactless identification using unique card IDs.
- Understood how to interface an LCD display, buzzer, LED indicators, and other electronic components.
- Improved skills in circuit design, debugging, and troubleshooting hardware and software issues.
- Learned to develop logic for student attendance tracking and entry/exit detection.
- Gained experience in system integration, testing, and collaborative project development.

Outcome

Outcome

- Developed a prototype of an RFID-Based Smart Attendance System to automate student identification and attendance tracking.
- Successfully implemented RFID card recognition using predefined student IDs.
- Enabled real-time display of student names and attendance status on an LCD.
- Integrated buzzer and LED indicators to provide immediate feedback during card scanning.
- Implemented entry/exit status tracking through repeated card scans.
- Created a low-cost, scalable attendance solution that can be enhanced with timestamp logging, data storage, and online monitoring in the future.

Future Scope

Enhancements

- Improved Data Storage: Store attendance records using an SD card or other external memory.
- Duplicate Scan Prevention: Introduce a short delay or scan validation mechanism to prevent accidental repeated scans.
- Enhanced User Interface: Improve LCD messages to display student names, entry/exit status, and timestamps.
- System Reliability: Add better error handling for invalid RFID cards and hardware communication failures.

Extensions

Cloud-Based Attendance: Integrate Wi-Fi using an ESP8266 or ESP32 to upload attendance records to an online database.

Mobile Application: Develop an application for teachers to view attendance records remotely.

Automated Reports: Generate daily, weekly, and monthly attendance summaries.

Multiple Classroom Support: Extend the system to manage multiple classrooms and larger student groups.

Low-Attendance Alerts: Notify teachers or parents when a student's attendance falls below a defined threshold.

Advanced Analytics: Analyze attendance patterns to identify frequent absences and improve monitoring.

THANK YOU